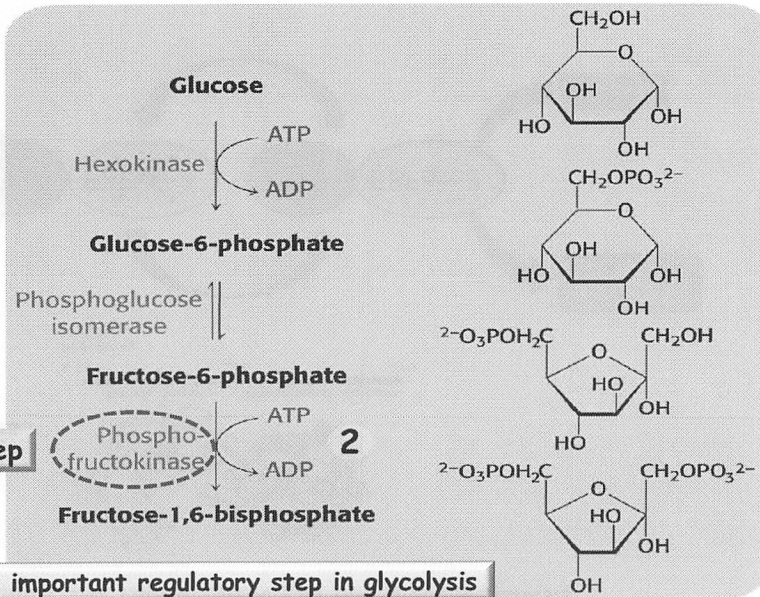


Priming stage uses 2mol of ATP to convert glucose to F-1,6-BP

Stage 1



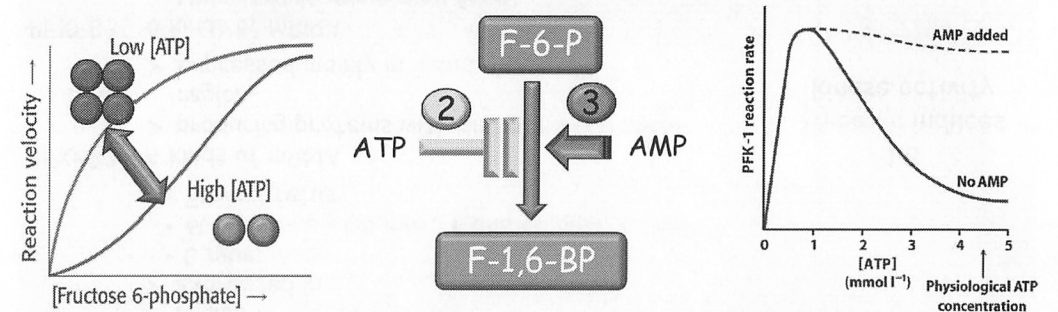
2nd irreversible step

PFK reaction is most important regulatory step in glycolysis

Active Phosphofructokinase(PFK) is a tetrameric enzyme

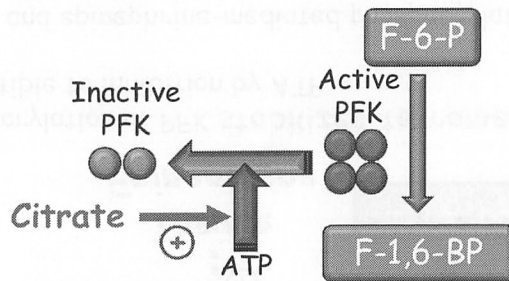
- ① PFK has three isoforms; PFK-L, -P (plasma), and -M.
- ② Skeletal muscle expresses only PFK-M, while other tissues do varying ratio of the isoforms.

ATP shows dual effects on PFK activity



- ① Each subunit has two ATP binding sites, a catalytic (binding constant: ~0.15 mM) and an inhibitory (binding constant: ~2.5 mM) site.
- ② PFK activity is activated at concentrations of 1 mM ATP (low ATP) and inhibited at higher concentrations (high ATP).
- ③ This inhibitory effect by ATP is relieved by AMP

Citrate enhances inhibitory effect of ATP, resulting in decreased PFK activity



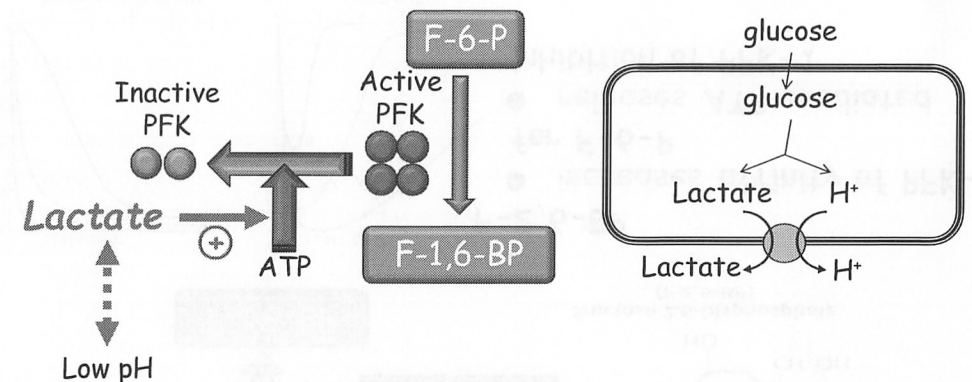
Citrate functions to preserve blood glucose

for tissues that are absolutely dependent on glucose as a fuel by inhibiting glycolytic pathway.

But, what energy status is it in the cells containing higher concentration of "citrate" ?

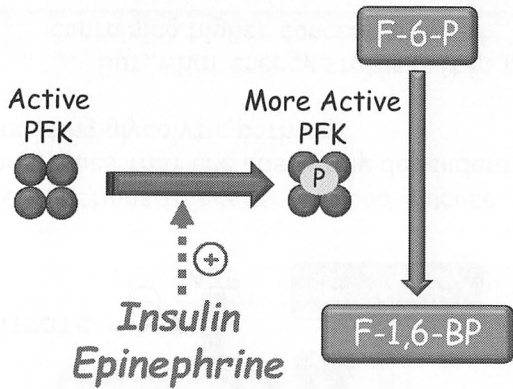
Citrate levels increase when *FAs* and *ketone bodies* are exclusively oxidized.

Low pH (acidosis) means accumulation of **lactate** in the cells, which can be reduced by inhibiting PFK activity.



Lactate is a *less* potent inhibitor for PFK activity compared to Citrate

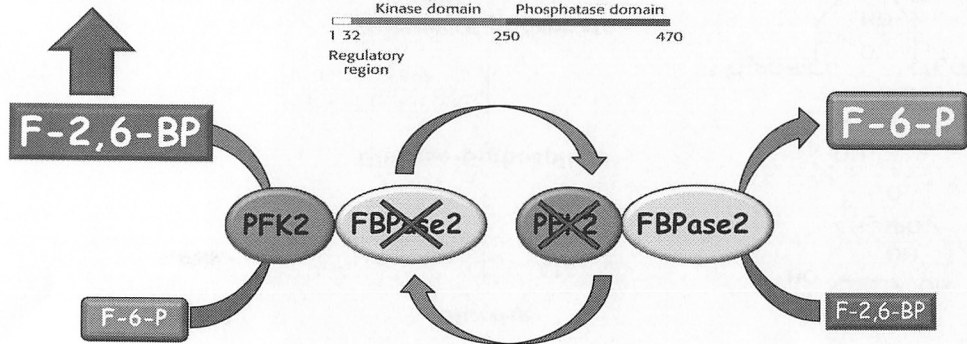
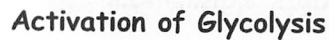
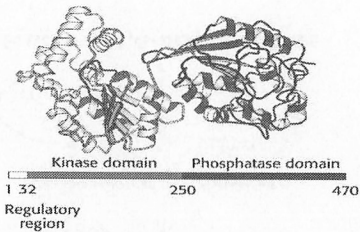
Hormonal regulation of PFK



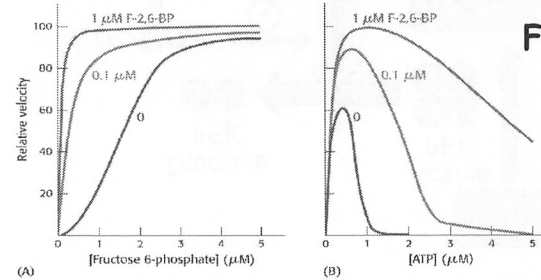
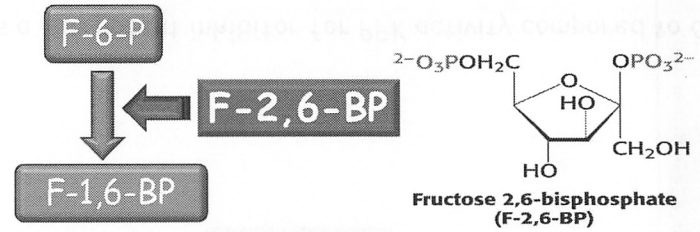
- Phosphorylation of PFK **stabilizes tetrameric form** with less susceptible to inhibition by ATP.
- Insulin and epinephrine-mediated phosphorylation of PFK increases its affinity for F-actin, by which stabilizes PFK tetrameric formation.

✓ In Type I-DM, phosphorylation of PFK at serine residues is reduced

PFK2/FBPase2 (PFKFB) is a homodimeric bifunctional enzyme to determine the flux of glycolysis and gluconeogenesis



Fructose-2,6-bisphosphate
is a most potent allosteric effector for PFK-1.



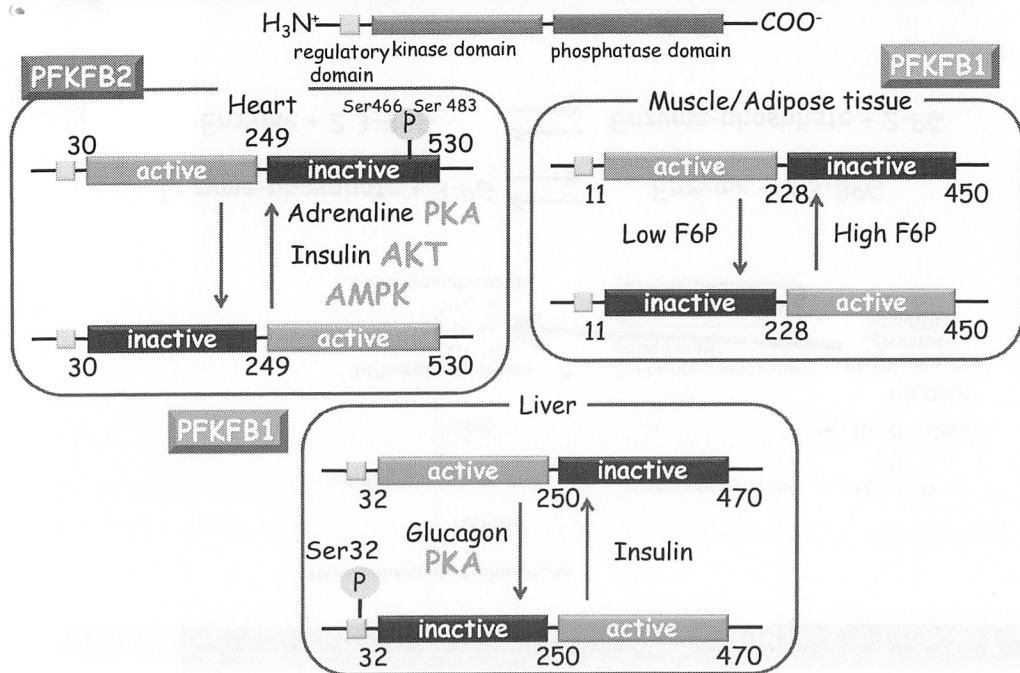
- increases affinity of PFK-1 for F-6-P
- releases ATP-mediated inhibition of PFK-1

- F-2,6-BP also stabilizes and promotes active tetramer formation of PFK-1.

Four isoforms of PFK2/FBPase2 with distinct properties

Gene Name	Expression	Kinase/Phosphatase activity
PFKFB1	3 kinds of mRNA ➤ producing proteins with distinct <i>N-terminal</i> region ➤ expressed in • L type; <i>liver</i> • M type; <i>skeletal muscle</i> and <i>adipose tissue</i> • F type; <i>fetus</i>	1.2
PFKFB2	4 kinds of mRNA ➤ producing proteins with distinct <i>C-terminal</i> region ➤ expressed mainly in <i>heart</i>	1.8 <i>Hypoxia</i> induces kinase activity
PFKFB3	6 kinds of mRNA ➤ Ubiquitously expressed form ➤ <i>Inducible</i> form (i-PFK2) ➤ <i>HIF-1 target gene</i>	> 700 <i>High kinase activity</i> <i>Hypoxia</i> induces kinase activity
PFKFB4	Several splicing variants expressed mainly in testis	0.9

Activity of PFK2/FBPase2 is regulated in a tissue-specific manner



Cancer and PFKFB3

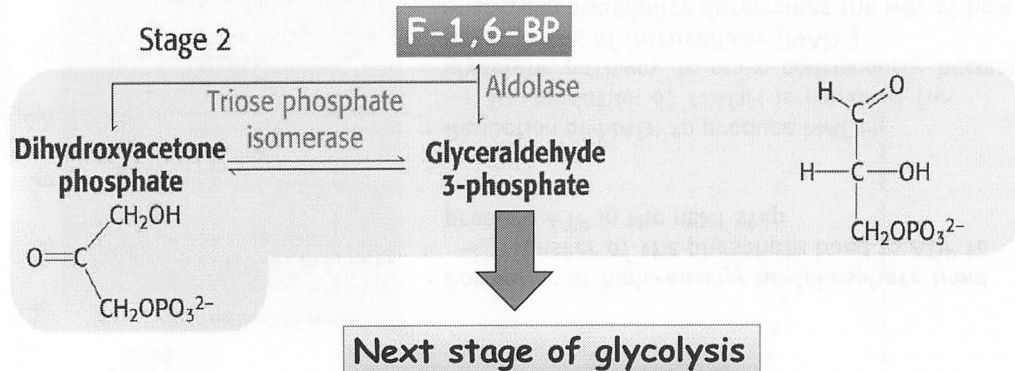
In cancer cells, glycolysis is active, in part, by stimulating a hypoxia sensitive transcription factor, HIF-1.

- HIF-1 mainly activates **PFKFB3** that shows **high kinase activity**.
- Phosphorylation at Ser461 by **AMPK** **activates** its kinase activity.
- Hypoxia also phosphorylates Ser461 of PFKFB3, resulting in increase in its kinase activity.

- **TIGAR** (Tumor protein 53-induced glycolysis and apoptosis regulator) is induced by **p53**.

- TIGAR resembles PFKFB in structure, but **lacks kinase activity**.
- Therefore, TIGAR **reduces F-2,6-BP levels** and inhibits glycolysis, by which cancer cells undergo apoptosis.
- p53 also activates a **copper transporter, SCO2** (synthesis of cytochrome oxidase-2), which is necessary for assembly of COX.

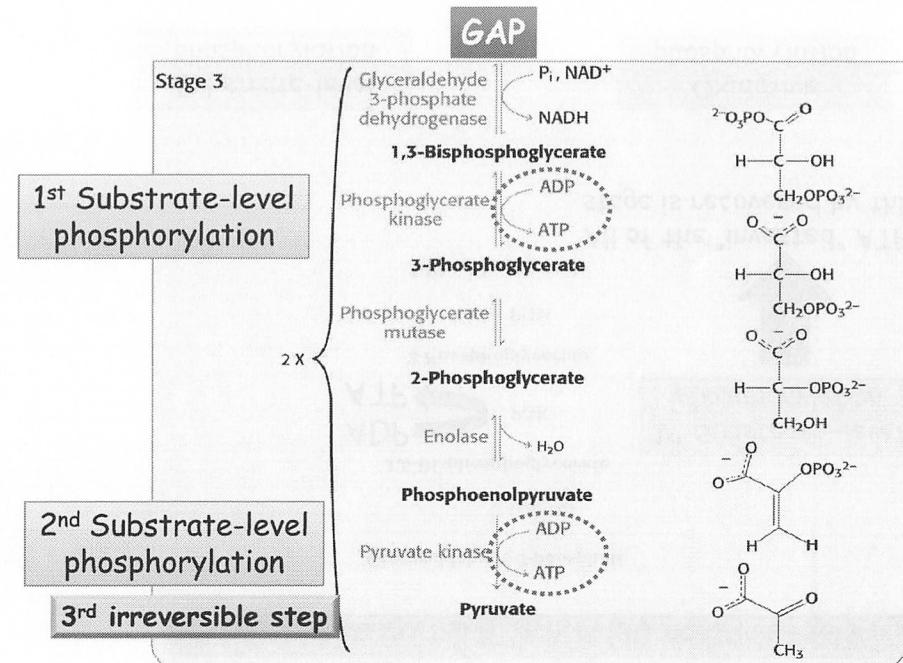
Splitting stage



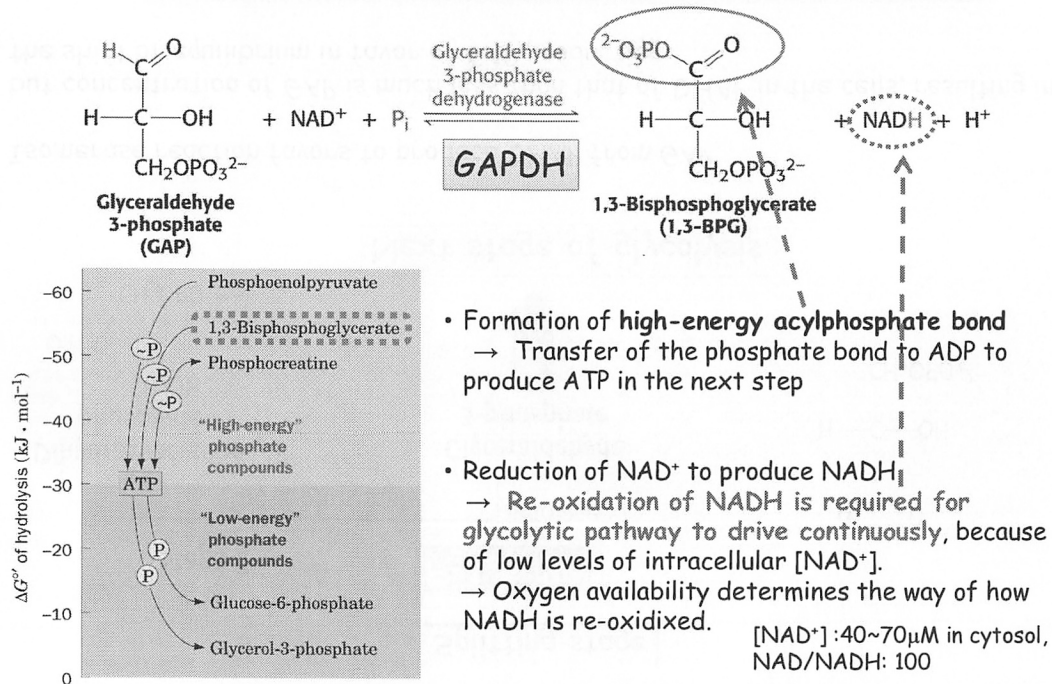
Isomerase reaction favors to produce DHAP from GAP,

but concentration of GAP is much less than that of DHAP in the cells, resulting in the shift of equilibrium in favor of GAP production

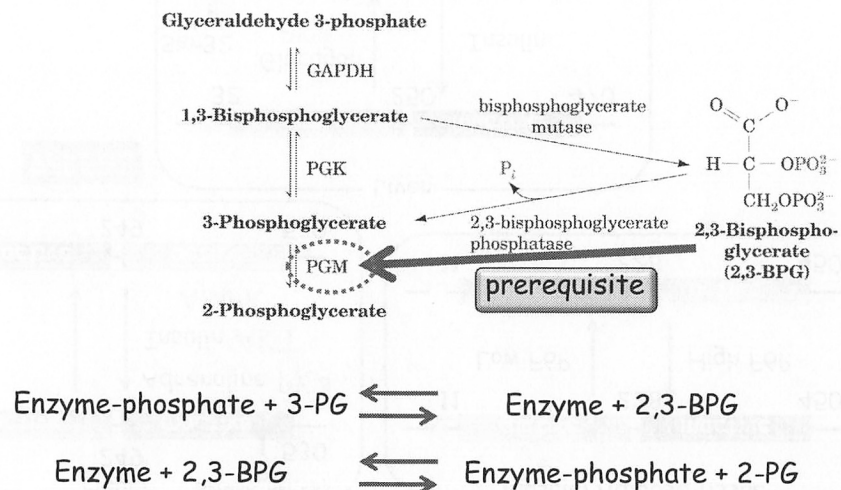
Energy salvaging stage, a final stage of glycolysis



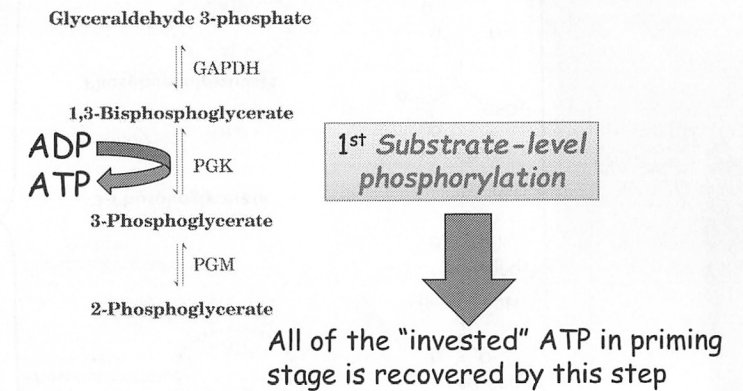
GAPDH produces high-energy phosphate compound, 1,3-BPG



2,3-BPG is a prerequisite for synthesis of 2-phosphoglycerate by PGM



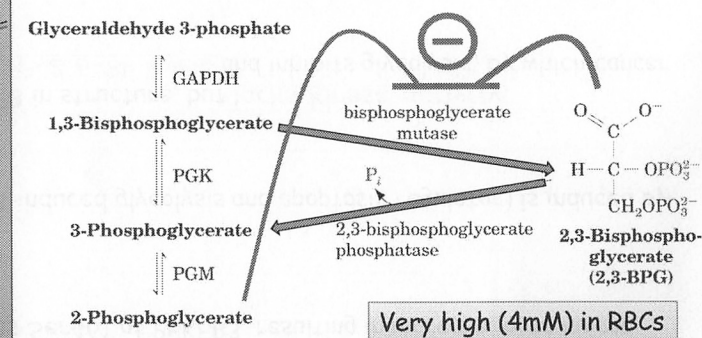
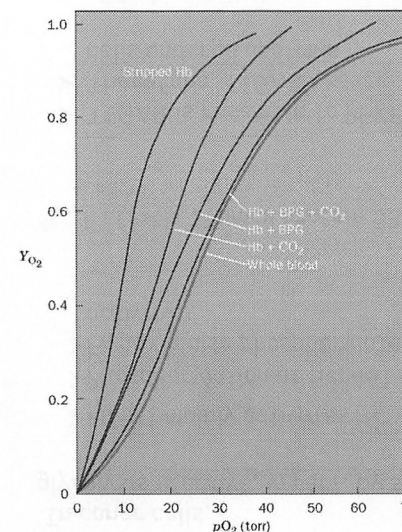
PGK catalyzes first ATP generating reaction in glycolysis



Substrate-level phosphorylation

Oxidative phosphorylation

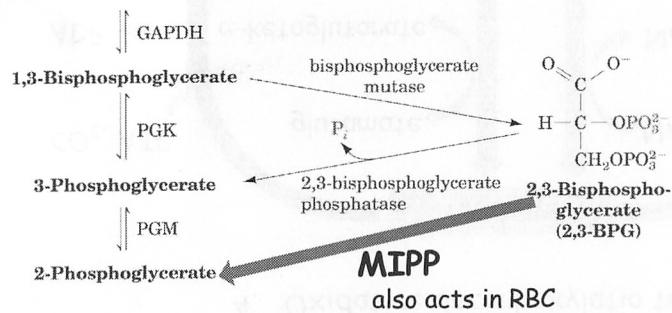
In red blood cells, a large portion (about 20%) of 1,3-BPG is metabolized to 2,3-BPG



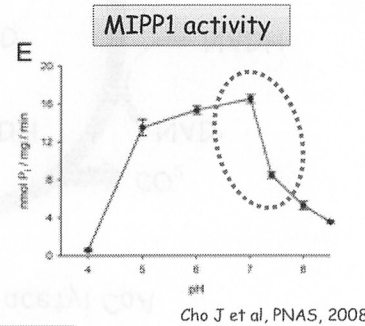
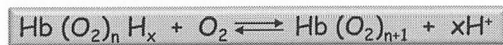
2,3-BPG lowers affinity of deoxyHb for oxygen, enabling to release greater amounts of oxygen in poorly oxygenized tissues.

In red blood cells, 2,3-BPG is also metabolized to 2-PG

Glyceraldehyde 3-phosphate



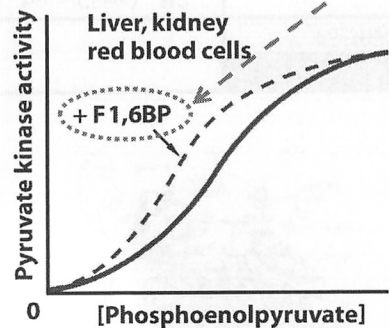
MIPP
also acts in RBC



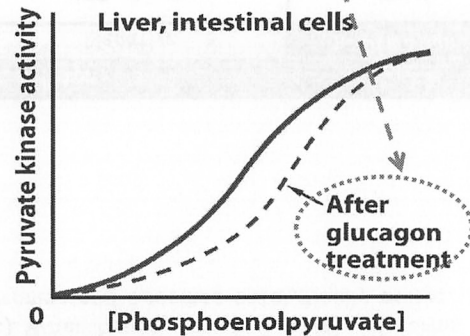
When Hb releases oxygen in peripheral tissues,
[H⁺] in RBCs decreases (Bohr's effect).
In this condition, MIPP activity decreases to greater extents,
resulting in an increase in 2,3-BPG and maintenance of oxygen delivery to the tissues.
This might serve as a feed-back mechanism against Bohr's effect

MIPP : multiple inositol polyphosphate phosphatase

Hepatic PK activity is regulated by allosteric effector and covalent modification



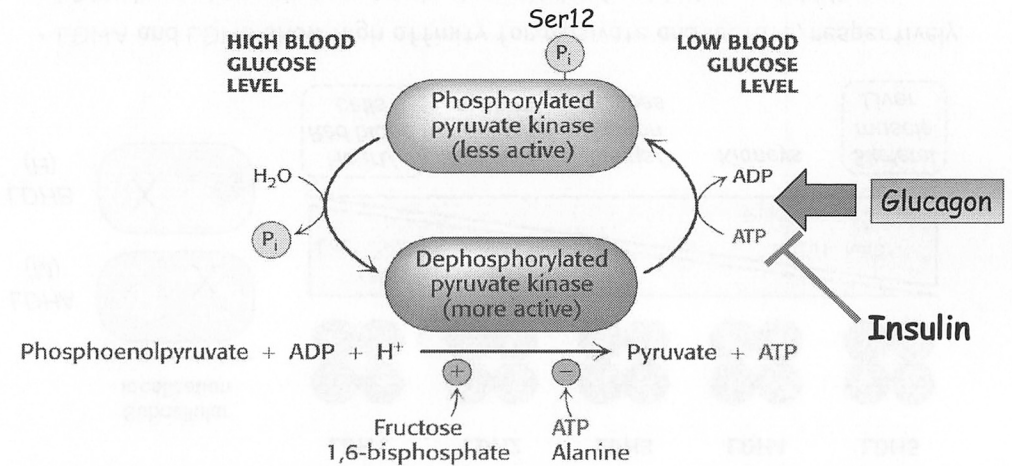
linkage of regulation of PK and that of PFK1



➤ Phosphorylation of PK by *PKA* decrease its activity in response to glucagon.

➤ Whereas *insulin* dephosphorylates and activates *PK*.

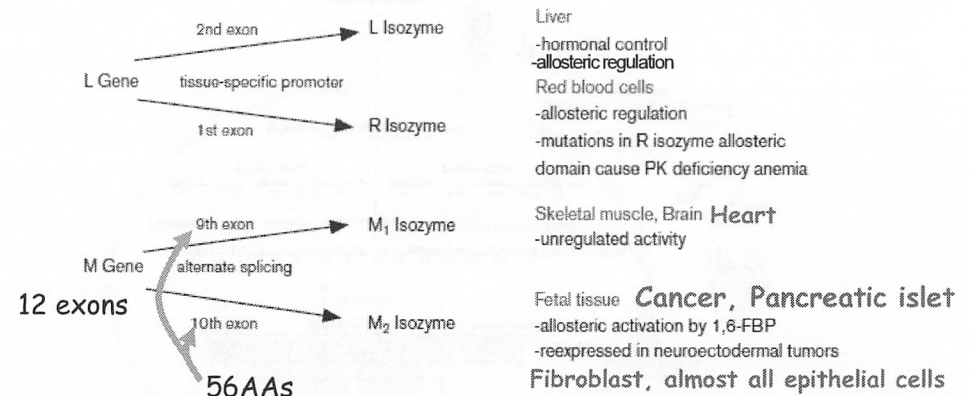
Pyruvate kinase is an enzyme catalyzing third irreversible step and 2nd substrate-level phosphorylation in the glycolysis



Hepatic PK expression is induced by insulin and high carbohydrate intake, whereas its expression is low under fasting and diabetic state

PK has four isoforms, *PKM1*, *PKM2*, *PKL* and *PKR*

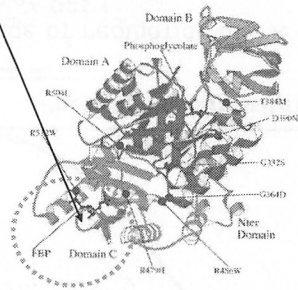
- PKLR encodes both PKL and PKR through use of alternative promoters and splicing.
- PKR is expressed only in RBC, while PKL is expressed in liver, kidney and small intestine.



➤ Liver and proximal tubules of kidney do not express PKM.

PK has four isoforms, *PKM1*, *PKM2*, *PKL* and *PKR*

- *PKR* forms homotetramer and contains four domains, domain A, B, C and a small N-terminal domain.
- Domain C is a binding site for F-1, 6-BP.

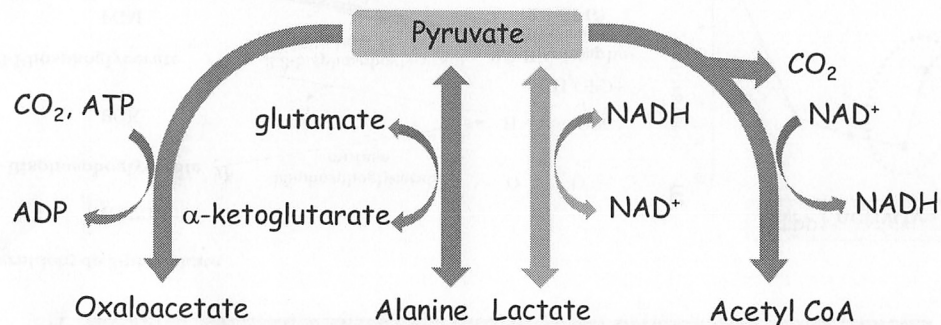


- In the absence of F-1, 6-BP, PK exist as a monomer (inactive). When bound with F-1,6-BP, PK forms homotetramer and becomes catalytically active.

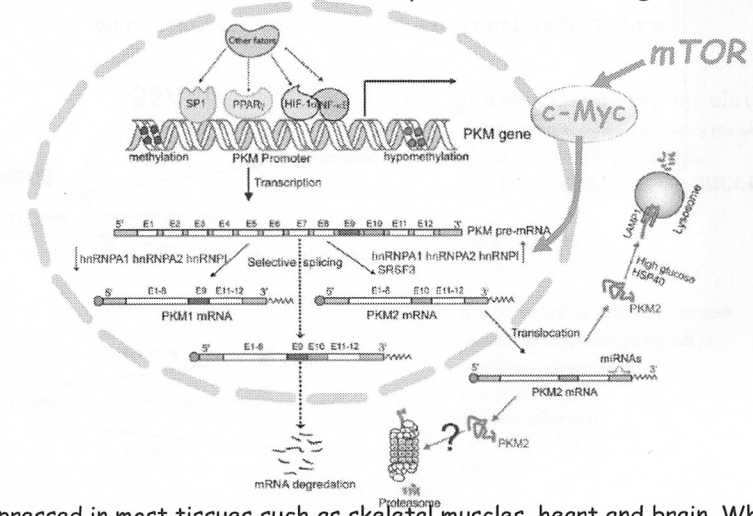
	Allosteric effector		Covalent modification
	Positive	Negative	
L type (liver), M2 type	F-1,6-BP	ATP, alanine	+ (PKA)
M1 type (Muscle, Brain)	×	×	-

Fate of pyruvate in mammals

1. Anaerobic reduction to lactate
2. Carboxylation to oxaloacetate
3. Transamination to alanine
4. Oxidative decarboxylation to acetyl CoA



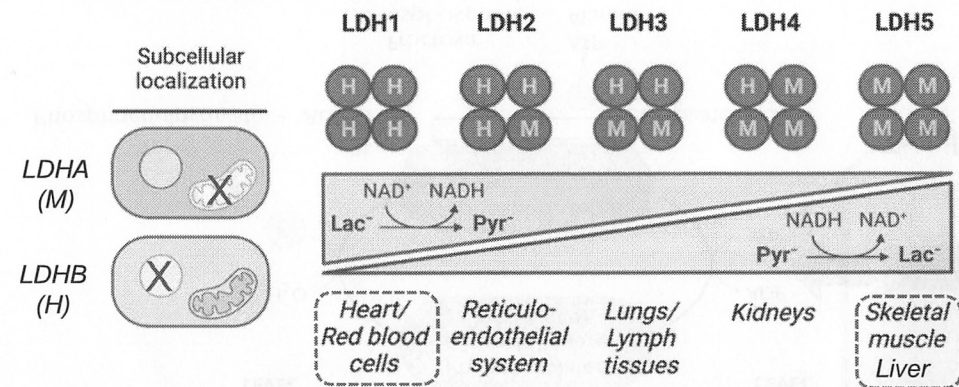
PKM1 and *PKM2* are alternatively transcribed using distinct exon



- *PKM1* is expressed in most tissues such as skeletal muscles, heart and brain. Whereas *PKM2* is present in embryonic tissues and cancer.
- *PKM1* forms a stable homotetramer like other PKs and shows high enzymatic activity.
- *PKM2* associates as either a tetramer (high activity) or a dimer (low activity). In some cancers, *PKM2* is highly expressed in a dimeric form.

Lactate dehydrogenase (LDH)

- Mammalian cells has two distinct types of LDH subunits, M type (LDHA) and H type (LDHB)
- LDH is a tetrameric enzyme



- LDHA and LDHB show high affinity for pyruvate and lactate, respectively.
- LDH plays a key role in not only the switch of oxidative metabolism to glycolysis but also gluconeogenesis.
- Plasma LDH serves as a biomarker of various pathological conditions such as liver damage, heart attack, viral infection, and cancer. (Normal range in serum: 200-400 IU/L)